Unreal Is Not Open:
prediction, horizons and the future in relativistic becoming

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Abstract

An event-relative interpretation of local relativistic becoming places what is real from an event between its chronological and causal past. Does the unreality of the future from an event make the future open from it? We separate the two questions. Call an event guaranteed from b if every past-inextendible causal curve from it meets b's chronological past; then every deterministic law with the domain-of-dependence property fixes it from what is real from b. The notion is essentially Geroch's domain of prediction. We characterize it by a condition on the slices of any Cauchy time function, show that it is conformally invariant, and compute it for the standard cosmological models. In Minkowski spacetime, spatially flat and open big-bang universes, the closed radiation universe and de Sitter spacetime, nothing outside an event's past is guaranteed. In Minkowski spacetime and the flat universes no local law need fix more than the causal past: a conformally coupled scalar field is fixed from an event exactly there. In the closed dust model everything is guaranteed from every event after maximum expansion, the known moment at which the particle horizon disappears. Before it, the same scalar field is already fixed on three diamonds: the event's past, an antipodal echo of it, and a region of the event's own timelike future just before the big crunch, while the intervening future stays open. For this field, which travels on light rays alone, what is fixed is decided by null geodesics rather than by all causal curves. Adding an arbitrarily small ordinary physical mass can remove exact fixing at a selected interior future event on the same dust background. Yet the hidden target differences, under a stated conformal-data norm bound, are $O(m^2)$: loss of exact fixing is distinct from practical unpredictability. Other local laws fix more: one with a preferred flow fixes everything even in Minkowski spacetime. So unreality does not entail openness. How much of the unreal future is fixed from an event depends on the spacetime's causal structure, whose guarantee can change with cosmic epoch, and on the fields, which that structure does not decide. A closing remark records what the corresponding standpoint logic does and does not show about cosmological horizons.

1 Introduction

Stein's theorem says that in Minkowski spacetime a transitive, symmetry-invariant relation of becoming that makes some past event definite is the causal order or trivial [6, 20, 21]. A companion paper [22] classifies becoming relations assigned uniformly to globally hyperbolic metric spacetimes, assuming chronological becoming, transitivity and preservation and reflection under causally convex isometric embeddings. In its nontrivial branch, what has become for an event b includes its chronological past and never goes beyond its causal past. The domain includes geometries not required to solve particular Einstein–matter equations; the locality premise is therefore a uniform metric-only requirement, not merely the absence of superluminal signalling. On views that read "has become for b" as "is real from b" [9, 15], the future of b is not real from b. A theory with an

additional clock or foliation and only embeddings preserving that structure changes these assumptions; it is not excluded simply by calling it nonlocal. The present paper takes the event-relative interpretation as a premise, not as an ontology derived from physics.

It is tempting to infer that the future is *open* from b. The inference fails, and the way it fails is instructive. Openness in the sense relevant here is a matter of what the real fixes, and that depends on the laws and on the global causal structure. The point is old in outline: a growing block [4] combined with deterministic laws makes the future unreal yet fixed, and Rakíc's sea-fight case turns on an event outside an observer's past that a decision inside it appears to settle [18]. We ask more precisely when, under specified relativistic laws and admissible solutions, what is real from an event fixes anything beyond itself. Geroch's domain of prediction supplies the tool [10], and Manchak's work on prediction its limits [16]. We use them to answer four questions:

1. Which events does the causal structure alone guarantee to be fixed, under any deterministic local law, by what is real from b (Theorem 3)?
2. In the standard cosmological models, how much is that, and how much more can a particular field fix (Section 3)?
3. Can changing the field equation change exact fixing without changing the causal structure, and what does that say about finite precision (Section 4)?
4. What follows for the relation between unreality and openness (Section 5)?

Conventions. Spacetimes are smooth, connected, time-oriented and globally hyperbolic. $I^-_{\mathrm{s}}(p)$ is the strict chronological past (via timelike curves), $J^-(p)$ the causal past; $x \leq y$ means $x \in J^-(y)$, and $x \ll y$ means $x \in I^-_{\mathrm{s}}(y)$. We use that $\leq$ is closed, that $J^-(x) \cap S$ is compact for every Cauchy surface S, push-up ($y \leq z \ll b$ implies $y \ll b$), and that $M = I^-_{\mathrm{s}}(S) \cup S \cup I^+_{\mathrm{s}}(S)$ for a Cauchy surface S [17]. A *Cauchy time function* is a continuous surjection $t : M \to (\alpha, \beta)$, $-\infty \leq \alpha < \beta \leq \infty$, strictly increasing along future-directed causal curves, whose level sets $S_c = t^{-1}(c)$ are Cauchy surfaces [5].

2 Guaranteed from an event

Definition 1. An event x is *guaranteed from* b if every past-inextendible causal curve from x meets $I^-_{\mathrm{s}}(b)$. Write $\mathrm{Det}(b)$ for the set of such events. Clearly $I^-_{\mathrm{s}}(b) \subseteq \mathrm{Det}(b)$.

$\mathrm{Det}(b)$ is a property of the spacetime and the event. To connect it with what the laws fix, let $\mathcal{L}$ be the set of solutions of some field equations on M. Say that x is *fixed from* b *under* $\mathcal{L}$, and write $x \in F_{\mathcal{L}}(b)$, if any two solutions in $\mathcal{L}$ that agree on $I^-_{\mathrm{s}}(b)$ agree at x. The equations, background and globally admissible solution class are held fixed in this comparison; they are inputs to the definition, not data an observer is assumed to have measured in the past. Here agreement means equality of the specified classical fields on the whole open region, not equality of a finite collection of detector records. Say that $\mathcal{L}$ has the *domain-of-dependence property* if, for every open $U \subseteq M$, any two solutions that agree on U agree at every event all of whose past-inextendible causal curves meet U. We assume this uniqueness property explicitly; it is not inferred merely from an operational no-signalling condition.

Remark 2 (What is guaranteed and what is fixed)**.** If $\mathcal{L}$ has the domain-of-dependence property and consists of continuous fields, then

$$\mathrm{Det}(b) \cup J^-(b) \subseteq F_{\mathcal{L}}(b),$$

since $J^-(b)$ is the closure of $I^-_{\rm s}(b)$ in a globally hyperbolic spacetime [17]. So $\mathrm{Det}(b)$ is fixed under every such law: it is what the causal structure guarantees, whatever the fields. The inclusion is in general strict, and nothing about $F_{\mathcal{L}}(b)$ beyond it follows from $\mathrm{Det}(b)$ being small. For example, on Minkowski spacetime with the unit timelike field $u = \partial_t$, the local equations $u \cdot \nabla Q = P$, $u \cdot \nabla P = 0$ have the solutions $Q = A(\vec{X}) + tB(\vec{X})$, $P = B(\vec{X})$ for arbitrary smooth A, B. The values of (Q, P) at one point of an integral curve fix them along it, and the past integral curve from x is a past-inextendible causal curve, so the law has the domain-of-dependence property; and every integral curve of u enters $I^-_{\rm s}(b)$, so $F_{\mathcal{L}}(b) = M$ for every b. A law of this kind uses background structure beyond the metric, but nothing in the domain-of-dependence property excludes it. Neither notion is Belnap's settledness, on which a proposition is settled at a point event if it holds in every possible history through it [2]; under indeterministic laws they come apart from it, and we say nothing about such laws. We use $I^-_{\rm s}(b)$, which every local becoming theory counts as real from b, rather than $J^-(b)$; the characterization below uses that $I^-_{\rm s}(b)$ is open.

Relation to Geroch. Geroch's domain of prediction $P(q)$ is the set of x such that every past-directed timelike curve from x without past endpoint enters $I^-(q)$, but $I^-(x) \not\subseteq I^-(q)$ [10, sec. 3]. $\mathrm{Det}(b)$ is the causal-curve analogue of the union of $P(b)$ with the events whose pasts lie in that of b; we expect the two to differ at most on null boundaries, but have not proved it. Geroch gives an equivalent form by achronal surfaces in $I^-(q)$ [10, sec. 3]; Theorem 3 is a slicing form for globally hyperbolic spacetimes.

Theorem 3. *Let $t : M \to (\alpha, \beta)$ be a Cauchy time function. Then $x \in \mathrm{Det}(b)$ if and only if there is $c \in (\alpha, t(x))$ with $J^-(x) \cap S_c \subseteq I^-_{\rm s}(b)$.*

Proof. ($\Leftarrow$) A past-inextendible causal curve from x crosses the Cauchy surface S_c, at a point of $J^-(x) \cap S_c \subseteq I^-_{\rm s}(b)$.

($\Rightarrow$) Suppose that for every $c \in (\alpha, t(x))$ the set $K_c = (J^-(x) \cap S_c) \setminus I^-_{\rm s}(b)$ is nonempty; it is compact, since $I^-_{\rm s}(b)$ is open. Fix $t(x) > c_1 > c_2 > \dots$ with $c_n \to \alpha$, and let

$$P_n = \{(z_1, \dots, z_n) \in K_{c_1} \times \cdots \times K_{c_n} : x \geq z_1 \geq z_2 \geq \cdots \geq z_n\}.$$

P_n is compact, since $\leq$ is closed. It is nonempty: pick $y \in K_{c_n}$ and a causal curve from x to y; it crosses each S_{c_i}, $i < n$, at some $z_i \leq x$, and $z_i \notin I^-_{\rm s}(b)$, for otherwise $y \leq z_i \ll b$ would give $y \ll b$. The maps $P_{n+1} \to P_n$ forgetting the last coordinate are continuous, so the inverse limit of the nonempty compact spaces P_n is nonempty, giving $x \geq z_1 \geq z_2 \geq \cdots$ with $z_n \in K_{c_n}$. Join consecutive terms by causal curves and concatenate. Each segment avoids $I^-_{\rm s}(b)$ by push-up, as before. Along the concatenation $t \to \alpha$, so it has no past endpoint in M: it is a past-inextendible causal curve from x missing $I^-_{\rm s}(b)$. □

Corollary 4. *If $I^-_{\rm s}(b)$ contains a Cauchy surface S, then $\mathrm{Det}(b) = M$.*

Proof. If $x \in I^-_{\rm s}(S) \cup S$, then $x \leq s \ll b$ for some $s \in S$, so $x \in I^-_{\rm s}(b)$. If $x \in I^+_{\rm s}(S)$, extend a past-inextendible causal curve from x to an inextendible one; it meets S, and not to the future of x, since then $s' \ll x \leq s$ for some $s, s' \in S$, contradicting achronality. So the curve from x meets $S \subseteq I^-_{\rm s}(b)$. □

Since $\mathrm{Det}(b)$ is defined by causal curves alone, it depends only on the conformal class of the metric; conformal diagrams decide it.

3 The standard models

Lemma 5. *Let (Σ, d) be $\mathbb{R}^3$, hyperbolic space H^3, or the unit sphere S^3 with $r_2 \le \pi$. Then the closed ball $\bar{B}(X, r_1)$ lies in the open ball $B(Y, r_2)$ if and only if $d(X, Y) + r_1 < r_2$.*

Proof. ($\Leftarrow$) is the triangle inequality. ($\Rightarrow$) Let $r_1 \ge 0$ and suppose $d(X, Y) + r_1 \ge r_2$. On $\mathbb{R}^3$ and H^3, where geodesics extend indefinitely and minimize, or on S^3 when $d(X, Y) + r_1 \le \pi$, follow a minimizing geodesic from Y through X (any geodesic from X if $X = Y$) a further distance r_1; the endpoint lies in $\bar{B}(X, r_1)$ at distance $d(X, Y) + r_1 \ge r_2$ from Y. If $d(X, Y) + r_1 > \pi$ on S^3, the antipode of Y is at distance $\pi - d(X, Y) < r_1$ from X and $\pi \ge r_2$ from Y. □

Proposition 6. *Let M be conformal to $(\eta_-, \eta_+) \times \Sigma$ with metric $-d\eta^2 + d\Sigma^2$, where Σ is $\mathbb{R}^3$, H^3, or the unit S^3 with $\eta_+ - \eta_- \le \pi$. Then $\mathrm{Det}(b) = I_{\mathrm{s}}^-(b)$ for every b.*

Proof. Use $t = \eta$. For $x = (\eta_x, X)$, $b = (\eta_b, Y)$: $x \ll b$ iff $d(X, Y) < \eta_b - \eta_x$ (on S^3, $\eta_b - \eta_x < \pi$). For $\eta_- < c < \eta_x$, $J^-(x) \cap S_c = \bar{B}(X, \eta_x - c)$ and $I_{\mathrm{s}}^-(b) \cap S_c = B(Y, \eta_b - c)$, with $\eta_b - c < \pi$ on S^3. By Lemma 5 the inclusion holds iff $d(X, Y) < \eta_b - \eta_x$, i.e. iff $x \ll b$. Apply Theorem 3. □

This covers Minkowski spacetime, spatially flat and open FLRW universes with a big bang (conformal to $\eta > 0$), the closed radiation universe ($a \propto \sin\eta$, $\eta \in (0, \pi)$) and global de Sitter spacetime ($\Sigma = S^3$, $\eta \in (-\pi/2, \pi/2)$) [14]. In none of them is anything outside an event's chronological past guaranteed from it. In Minkowski spacetime and the spatially flat universes this bound is attained by a natural field.

Proposition 7. *Let $M = (\eta_-, \eta_+) \times \mathbb{R}^3$ with $g = a(\eta)^2(-d\eta^2 + d\vec{X}^2)$, $a > 0$ smooth, and let $\mathcal{L}$ be the smooth solutions of the conformally coupled scalar equation $\Box_g \phi - \frac{1}{6} R\phi = 0$. Then $F_{\mathcal{L}}(b) = J^-(b)$ for every b.*

Proof. $J^-(b) \subseteq F_{\mathcal{L}}(b)$ by continuity, as in Remark 2. Conversely, ϕ is a solution iff $\psi = a\phi$ solves the flat wave equation $\partial_\eta^2 \psi = \Delta\psi$ on M [23, app. D]. Let $b = (\eta_b, \vec{Y})$ and $x = (\eta_x, \vec{X}) \notin J^-(b)$, so $\eta_x - \eta_b + |\vec{X} - \vec{Y}| > 0$, since the causal curves of M are those of the conformal Minkowski metric and the slab is causally convex in Minkowski spacetime. Choose a unit vector $\hat{k}$ with $\hat{k} \cdot (\vec{X} - \vec{Y}) = -|\vec{X} - \vec{Y}|$, a smooth f vanishing on $(-\infty, 0]$ and positive on $(0, \infty)$, and put $\psi = f(u)$ with $u = \eta - \eta_b - \hat{k} \cdot (\vec{Z} - \vec{Y})$ at $(\eta, \vec{Z})$. This solves the wave equation. On $I_{\mathrm{s}}^-(b)$, $\eta - \eta_b < -|\vec{Z} - \vec{Y}|$, so $u < 0$ and $\psi = 0$; at x, $u > 0$ and $\psi \ne 0$. So ψ/a and 0 are solutions that agree on $I_{\mathrm{s}}^-(b)$ but not at x. □

Remark 8 (The class matters)**.** The witness in this proof is a plane wave, which has infinite energy. The verdict $F_{\mathcal{L}}(b) = J^-(b)$ belongs to the class of all smooth solutions as well as to the equation. The next proposition shows the verdict changing with the class for one equation on one spacetime.

Proposition 9 (Massive fields on Minkowski spacetime)**.** *On Minkowski spacetime $\mathbb{R}^{1+n}$ let $m > 0$ and consider real solutions of $\partial_t^2 \phi - \Delta\phi + m^2\phi = 0$.*

(a) For the class of all smooth solutions, $F(b) = J^-(b)$ for every b.

(b) For the class of finite-energy solutions (Cauchy data in $H^1 \times L^2$), $F(b)$ is the whole spacetime for every b, and the same holds with the strict chronological future of b as base.

Proof. (a) $J^-(b) \subseteq F(b)$ by continuity. Let $x \notin J^-(b)$, with $b = (t_b, \vec{Y})$. As in the proof of Proposition 7 choose a unit vector $\hat{k}$ such that $u = t - t_b - \hat{k} \cdot (\vec{Z} - \vec{Y})$ is negative on $I_{\mathrm{s}}^-(b)$ and positive at x, and put $v = t - t_b + \hat{k} \cdot (\vec{Z} - \vec{Y})$. For functions of (u, v) the equation reads

$4\phi_{uv} + m^2\phi = 0$. The function $j(z) = J_0(m\sqrt{z})$ is entire and satisfies $zj'' + j' + \frac{m^2}{4}j = 0$, so $R = j((u-s)(v-c))$ satisfies $R_{uv} = -\frac{m^2}{4}R$ for all real s, c. Let $h \geq 0$ be smooth with support in $(0, \infty)$ and put

$$\phi(u,v) = \int_0^u h(s)\, j\big((u-s)(v-c)\big)\, ds \quad (u > 0), \qquad \phi = 0 \quad (u \leq 0).$$

Since $R = 1$ at $s = u$, $\phi_u = h(u) + \int_0^u hR_u\, ds$, and so $\phi_{uv} = \int_0^u hR_{uv}\, ds = -\frac{m^2}{4}\phi$. Thus ϕ is a smooth solution vanishing on $I_{\mathrm{s}}^-(b)$. Taking c close to $v(x)$ and h a bump supported below $u(x)$ gives $\phi(x) > 0$.

(b) Take b at the origin and let ϕ have finite energy and vanish on $I_{\mathrm{s}}^-(b) = \{t < -|x|\}$. At each $t < 0$ the energy density vanishes on $|x| < |t|$, so all of the conserved energy lies in $|x| \geq |t|$. For a finite-energy solution that portion tends to zero as $t \to -\infty$: for $n = 1$ by Burnol [26, Theorem 1, with time reversed], and for every n by Yang [25, Remark 1.10], who states it for $m = 1$, to which the general case reduces by scaling. Hence the energy is zero, and $\phi = 0$ because $m > 0$. The future base is the time reverse. □

The solutions used in (a) grow like a modified Bessel function where $(u-s)(v-c) < 0$ and have infinite energy. The argument for (b) uses only the decay of energy outside the light cone, which fails at zero mass. A massive field carries nothing in from null infinity: its energy arrives from past timelike infinity, which lies in the closure of every past light cone. So on this background the past of an event fails to fix a finite-energy field only when the field is massless. Weinstein argues that prediction from a past light cone in electromagnetism needs the added assumption of no incoming radiation [27]; part (b) says that for a massive field of finite energy the corresponding statement is a theorem, and a time-symmetric one. The fixing in (b) uses the arbitrarily remote past, so it does not make such a field practically predictable (compare Section 4).

Proposition 10. *Let M be conformal to $(\eta_-, \eta_+) \times S^3$ with $\eta_+ - \eta_- > \pi$. Then* $\mathrm{Det}(b) = I_{\mathrm{s}}^-(b)$ *if $\eta_b \leq \eta_- + \pi$, and* $\mathrm{Det}(b) = M$ *if $\eta_b > \eta_- + \pi$. In the Einstein static universe ($\eta_\pm = \pm\infty$),* $\mathrm{Det}(b) = M$ *for every b.*

Proof. If $\eta_b > \eta_- + \pi$, take $c \in (\eta_-, \eta_b - \pi)$: $I_{\mathrm{s}}^-(b) \cap S_c$ is an open ball of radius greater than π, hence all of S_c, and Corollary 4 applies. If $\eta_b \leq \eta_- + \pi$, every $c > \eta_-$ has $\eta_b - c < \pi$, and the proof of Proposition 6 applies. □

For the closed dust FLRW model, $a(\eta) = \frac{1}{2}a_m(1-\cos\eta)$ with $\eta \in (0, 2\pi)$, the threshold is $\eta = \pi$, the moment of maximum expansion. That the particle horizon of the closed dust model disappears at maximum expansion, so that an observer can then in principle see the whole universe, is known [3, 12]; the Einstein static case is Geroch's [10, Fig. 5]. What Proposition 10 adds is the consequence for determination: with Corollary 4, once the particle horizon has vanished ($\eta_b > \pi$), everything is guaranteed from b, so that every law with the domain-of-dependence property fixes everything from what is real from b. This fits Manchak's theorem that if some event of b's timelike future lies in Geroch's domain of prediction of b, the spacetime admits a compact spacelike slice [16, p. 319]. That theorem concerns the geometric notion. The flow law of Remark 2 shows that it has no analogue for what a particular law fixes: in Minkowski spacetime, which has no compact slice, that law fixes all of b's timelike future. Manchak also records the view that "Maxwell's equations constrain some spacetimes in such a way as to allow for prediction" [16, p. 321 and n. 15], so the law-relative notion has antecedents. Earman's discussion explicitly includes the Cauchy-surface exception and an electromagnetic-constraint example citing Ellis and Sciama [8, pp. 125–130]. We have checked these selected original Earman pages; Ellis and Sciama (1972, pp. 41–42) remain unread here and

enter this comparison through Earman's discussion. The model-specific characterization below may overlap with those sources; we make no priority claim for it.

Before maximum expansion a particular field can fix more than is guaranteed. Light leaving a point of the closed model refocuses at its antipode after conformal time π, and the rescaled conformal scalar $\psi = a\phi$ returns there with its sign reversed.

Proposition 11. *Let M be the closed dust model, $g = a(\eta)^2(-d\eta^2 + d\Omega_3^2)$ with $a = \frac{1}{2}a_m(1-\cos\eta)$, $\eta \in (0, 2\pi)$, and $\mathcal{L}$ as in Proposition 7. Let $A(\eta, X) = (\eta + \pi, -X)$, where $-X$ is the antipode of X. Then $J^-(b) \cup \big(A(J^-(b)) \cap M\big) \subseteq F_{\mathcal{L}}(b)$. The set $A(J^-(b)) \cap M$ is disjoint from $I_{\mathrm{s}}^+(b)$, and if $\eta_b \le \pi$ it is nonempty and disjoint from* $\mathrm{Det}(b)$.

Proof. Here $\psi = a\phi$ solves the conformally invariant equation of the Einstein static universe, $\partial_\eta^2\psi = \Delta\psi - \psi$ on the unit S^3 [23, app. D], and by global well-posedness extends uniquely to $\mathbb{R}\times S^3$. Expanding in spherical harmonics of degree n, which have frequency $n+1$ and parity $(-1)^n$ under the antipodal map, each mode, and so every smooth solution, satisfies $\psi(\eta+\pi, -X) = -\psi(\eta, X)$. Two solutions that agree on $I_{\mathrm{s}}^-(b)$ agree, by continuity, on $J^-(b)$, and hence on $A(J^-(b)) \cap M$. If $(\eta, X) \in J^-(b)$ and $\eta + \pi < 2\pi$, then $d(X, Y) \le \eta_b - \eta$, so $d(-X, Y) = \pi - d(X, Y) \ge (\eta + \pi) - \eta_b$; thus $A(\eta, X) \notin I_{\mathrm{s}}^+(b)$. If $\eta_b \le \pi$, then $\mathrm{Det}(b) = I_{\mathrm{s}}^-(b)$ by Proposition 10, which lies in $\{\eta < \eta_b\}$, while $A(J^-(b))$ lies in $\{\eta > \pi\}$ and meets M, for instance at $A(\eta_b/2, Y)$ when $b = (\eta_b, Y)$. □

For $\eta_b \le \pi$, the echo events are unreal from b and fixed from b, but they lie beside b or on its future light cone, not in its timelike future. The field fixes more than the echo, including part of b's timelike future. Let $\Diamond_b = \{(\eta, X) \in \mathbb{R}\times S^3 : d(X, Y) + |\eta| < \eta_b\}$, the double cone in the Einstein static universe between b and its mirror image $(-\eta_b, Y)$, and $U_b = \Diamond_b \cup A\Diamond_b \cup A^2\Diamond_b$.

Theorem 12. *In the setting of Proposition 11, let $0 < \eta_b < \pi$. Then $F_{\mathcal{L}}(b) = M \cap \overline{U_b}$. In particular $F_{\mathcal{L}}(b)$ contains the nonempty set $\{(\eta, X) \in M : d(X, Y) \le \eta + \eta_b - 2\pi\}$, which lies in $I_{\mathrm{s}}^+(b)$.*

Proof. For data (f, g) at $\eta = 0$, the Einstein static solution is $\psi(\eta, X) = \partial_\eta[\sin\eta\, M_f(X, \eta)] + \sin\eta\, M_g(X, \eta)$, where $M_h(X, r)$ is the mean of h over the geodesic sphere of radius r about X; one checks this on each mode, using that the zonal function of degree n on S^3 is $\sin((n+1)r)/((n+1)\sin r)$. So $\psi(x)$ depends on the data only near the sphere $\Sigma_x = S(X, \eta_x)$, which is the point $-X$ when $\eta_x = \pi$.

Sufficiency. Solutions agreeing on $I_{\mathrm{s}}^-(b)$ agree on the ball $B(Y, \eta_b - \epsilon)$ in the slice $\eta = \epsilon$, hence on its domain of dependence in the Einstein static universe; the union over $\epsilon > 0$ is $\Diamond_b$. The identities $\psi \circ A = -\psi$ and $\psi \circ A^2 = \psi$ carry the agreement to $A\Diamond_b$ and $A^2\Diamond_b$. Moreover, $F_{\mathcal{L}}(b)$ is relatively closed in M: it is the intersection of the zero sets of all smooth difference solutions vanishing on $I_{\mathrm{s}}^-(b)$. Since M is open in the Einstein cylinder, taking the relative closure gives $M \cap \overline{U_b} \subseteq F_{\mathcal{L}}(b)$.

Necessity. Let $x \in M \setminus \overline{U_b}$. Replacing x by $A^{-1}x$ if $\eta_x > \pi$, assume $\eta_x \le \pi$. Let $C = \bar{B}(Y, \eta_b)$. Data supported off C give a solution vanishing on $I_{\mathrm{s}}^-(b)$: if $z \in I_{\mathrm{s}}^-(b)$ lay in its support, then $w \le z \ll b$ for some w in the support of the data, so $d(w, Y) < \eta_b$. If $\eta_x = \pi$, then $d(-X, Y) > \eta_b$, and data $(f, 0)$ with f a bump at $-X$ off C give $\psi(x) = -f(-X) \ne 0$. If $\eta_x < \pi$, then $\Sigma_x \not\subseteq C$: otherwise the connected set $S^3 \setminus C$ lies in one of the two open balls bounded by Σ_x, so the other closed ball lies in C, which by the argument of Lemma 5 means $d(X, Y) + \eta_x \le \eta_b$ or $d(-X, Y) + \pi - \eta_x \le \eta_b$, that is, $x \in \overline{\Diamond_b} \cup \overline{A\Diamond_b}$. Take $p \in \Sigma_x \setminus C$ and data $(0, g)$ with $g \ge 0$ supported near p off C and $g(p) > 0$; then $\psi(x) = \sin\eta_x\, M_g(X, \eta_x) > 0$.

Finally, if $d(X, Y) \le \eta + \eta_b - 2\pi$, then $d(X, Y) < \eta - \eta_b$ because $\eta_b < \pi$, so $(\eta, X) \in I_{\mathrm{s}}^+(b)$; the set contains (η, Y) for $\eta \ge 2\pi - \eta_b$. □

On b's comoving geodesic $X = Y$, the field is fixed from b exactly for $\eta \leq \eta_b$ and for $\eta \geq 2\pi - \eta_b$, and open in between: the end is fixed before the middle. The reason is that the past light cone of an event just before the crunch wraps around the universe and refocuses near Y at the big bang, inside what is real from b. The same sphere argument shows that, up to boundary, $M \cap U_b$ is the set of events all of whose past-inextendible null geodesics meet $I_{\mathrm{s}}^-(b)$, so for this field, which propagates on light rays alone, the null geodesics decide what is fixed. The same argument applies to global de Sitter spacetime, conformal to the slab $(-\pi/2, \pi/2)$, where the field is fixed from b on the antipodal comoving geodesic for all $\eta > -\eta_b$, although no signal from that geodesic ever reaches the comoving geodesic of b. So the geometric threshold at maximum expansion marks when everything is guaranteed, not when this field first fixes anything beyond the past.

For a law whose fixed set itself changes at the threshold, take the comoving flow $u = a^{-1}\partial_\eta$ of the closed dust model with the equations of Remark 2; here u is the dust's own four-velocity, so the law adds no structure the model lacks. The comoving line through X meets $I_{\mathrm{s}}^-(b)$ iff $d(X, Y) < \eta_b$, so, by continuity in X and by choosing data supported off the closed ball, $F_{\mathcal{L}}(b) = (0, 2\pi) \times \bar{B}(Y, \eta_b)$. This is a proper subset of M, containing events in b's timelike future, if $\eta_b < \pi$, and all of M if $\eta_b \geq \pi$. Under this law the answer to "is the future fixed from here?" differs between events before and after maximum expansion.

Whether in the closed models some local law fixes no more than $\mathrm{Det}(b) \cup J^-(b)$ we leave open.

4 Changing the law and changing the records

The massless example is special. Its extra fixing can change even when the metric and its causal curves are held fixed. Let $\mathcal{L}_m$ consist of all smooth real solutions on the closed dust strip of

$$(\Box_g - R_g/6 - m^2)\phi = 0,$$

with ordinary constant physical mass $m \geq 0$, and write $F_m(b)$ for its fixing set. Set $\Lambda^2 = -\Delta_{S^3} + 1$ and $T = 2\pi$. With $\psi = a\phi$ and $\lambda = m^2$, the rescaled equation is

$$\psi_{\lambda,\eta\eta} + (\Lambda^2 + \lambda q(\eta))\psi_\lambda = 0, \qquad q(\eta) = a(\eta)^2. \tag{1}$$

In particular a constant addition to the rescaled potential would be a different perturbation, corresponding to physical potential δ/a^2.

Proposition 13 (Loss of exact fixing under ordinary mass)**.** *Fix the dust amplitude $a_m > 0$. For each $b = (\tau, Y)$ with $0 < \tau < \pi$, there are a fixed event $x_* = (T - s, Y)$, $0 < s < \tau$, and $m_* > 0$ such that*

$$x_* \in F_0(b) \cap I_{\mathrm{s}}^+(b), \qquad x_* \notin F_m(b) \quad \textit{for every } 0 < m < m_*.$$

The background metric is the same for all these solution classes.

Proof. Choose a nonzero smooth $v \geq 0$ supported strictly outside $\bar{B}(Y, \tau)$, and solve (1) with data $(0, v)$ at the auxiliary cylinder slice $\eta = 0$. The coefficient $q = a^2$ is smooth through 0 and T, so ordinary smooth Cauchy evolution and parameter dependence apply on the extended cylinder. This is a device for constructing solutions on the open physical strip, not a physical continuation through its singularities. For $z = (\eta, Z) \in I_{\mathrm{s}}^-(b)$, the initial ball $\bar{B}(Z, \eta)$ lies in $B(Y, \tau)$ by the triangle inequality. Finite propagation therefore gives $\psi_\lambda = 0$ throughout $I_{\mathrm{s}}^-(b)$ for every λ; the zero-order term does not change this conclusion.

Put $S_0(u) = \sin(u\Lambda)/\Lambda$. The first variation $w = \partial_\lambda \psi_\lambda|_{\lambda=0}$ has zero initial data and source $-q(u)S_0(u)v$. Since $S_0(T-u) = -S_0(u)$, Duhamel's formula gives

$$w(T) = \int_0^T q(u) S_0(u)^2 v \, du.$$

The frequencies are $k = n+1$. Expanding $q = (a_m^2/4)(3/2 - 2\cos u + \frac{1}{2}\cos 2u)$ gives

$$\int_0^T q(u) \frac{\sin^2(ku)}{k^2}\, du = \begin{cases} 5\pi a_m^2/16, & k = 1, \\ 3\pi a_m^2/(8k^2), & k \geq 2. \end{cases}$$

Thus, with Π_0 the projection onto spatial constants,

$$w(T) = \left(\frac{3\pi a_m^2}{8} \Lambda^{-2} - \frac{\pi a_m^2}{16} \Pi_0 \right) v.$$

The kernel of Λ^{-2} on the unit S^3 is $G(r) = (\pi - r)/(4\pi^2 \sin r)$: it solves the homogeneous radial equation away from its pole, is smooth at the antipode, and has the normalized singularity $1/(4\pi r)$. Since Π_0 has kernel $1/(2\pi^2)$, the kernel of the operator displayed above is

$$K(r) = \frac{a_m^2}{32\pi} \left(\frac{3(\pi - r)}{\sin r} - 1 \right) > 0 \qquad (0 < r \leq \pi),$$

using $(\pi - r)/\sin r \geq 1$, with its continuous value at $r = \pi$. Consequently $w(T, Y) > 0$. Continuity now permits a choice of a fixed $s \in (0, \tau)$ with $w(T - s, Y) > 0$. Theorem 12 puts $x_* = (T - s, Y)$ in $F_0(b) \cap I_{\mathrm{s}}^+(b)$, so $\psi_0(x_*) = 0$. Smooth dependence on λ yields

$$\psi_{m^2}(x_*) = m^2 w(x_*) + o(m^2) > 0$$

for every sufficiently small positive m. Because $a(x_*) > 0$, the physical field and zero agree on $I_{\mathrm{s}}^-(b)$ but differ at x_*. The event and data were chosen before the mass interval. □

This proves neither that all extra fixing is lost nor a formula for F_m. The mass threshold can depend on the target and data. The result concerns a linear classical field on a fixed background, without interactions, quantization or gravitational backreaction.

4.1 A norm bound for hidden alternatives

Exact fixing quantifies over every admissible difference, however small. To ask about its size, a restriction on alternatives must be supplied. Use the conformal Cauchy-data norm

$$N(f, v)^2 = \|\Lambda f\|_2^2 + \|v\|_2^2$$

at the auxiliary slice $\eta = 0$, with unit-S^3 volume $2\pi^2$. This is not an observer's measured physical-energy budget.

Proposition 14. *Let ψ_λ and ψ_0 have the same smooth data (f, v) and solve the cylinder equation with spatially homogeneous smooth real potential $\lambda q(\eta)$ and with zero potential, respectively. For $\eta \geq 0$, put $Q(\eta) = \int_0^\eta |q(u)|\, du$. Then*

$$\sup_X |\psi_\lambda(\eta, X) - \psi_0(\eta, X)| \leq \frac{|\lambda| Q(\eta) e^{|\lambda| Q(\eta)}}{\sqrt{6}}\, N(f, v). \tag{2}$$

If $f = 0$, the denominator improves to $\sqrt{12}$.

Proof. Let $C_{\lambda,k}$ and $S_{\lambda,k}$ be the fundamental solutions with data $(1,0)$ and $(0,1)$ at frequency $k \geq 1$. Their Volterra equations and Gronwall's inequality imply $|C_{\lambda,k}| \leq e^{|\lambda|Q/k}$ and $|S_{\lambda,k}| \leq e^{|\lambda|Q/k}/k$. Substitution back into those equations bounds the differences from the free multipliers by

$$|C_{\lambda,k} - \cos k\eta| \leq B/k, \qquad |S_{\lambda,k} - \sin(k\eta)/k| \leq B/k^2, \quad B = |\lambda|Qe^{|\lambda|Q}.$$

Acting on Λf rather than f, the first multiplier acquires another $1/k$. The addition identity for an orthonormal harmonic basis is $\sum_j |Y_{kj}(X)|^2 = k^2/(2\pi^2)$, since the eigenspace has dimension k^2 and its diagonal projector is rotation-invariant. Joint Cauchy–Schwarz on the coefficients of $(\Lambda f, v)$ bounds the squared evaluation norm by

$$\sum_{k=1}^{\infty} \frac{k^2}{2\pi^2} \left(\frac{B^2}{k^4} + \frac{B^2}{k^4} \right) = \frac{B^2}{6}.$$

Removing the position-data term halves this bound. The argument concerns smooth solutions and the bounded difference functional; it does not assign point values to arbitrary individual finite-energy fields. □

Apply this to any rescaled difference $h_m = a(\phi_1 - \phi_2)$ of two massive solutions agreeing exactly on $I_{\mathrm{s}}^-(b)$, with data (f, v). Smooth extension to zero implies that both data vanish on $B(Y, \tau)$. The massless solution h_0 with the same data vanishes on $I_{\mathrm{s}}^-(b)$ by finite propagation. Thus $h_0(x) = 0$ for every $x \in F_0(b)$, and (2) gives

$$|h_m(\eta, X)| \leq \frac{m^2 Q(\eta) e^{m^2 Q(\eta)}}{\sqrt{6}} N(f, v), \qquad (\eta, X) \in F_0(b), \tag{3}$$

where, for dust,

$$Q(\eta) = \frac{a_m^2}{4} \left(\frac{3\eta}{2} - 2\sin\eta + \frac{\sin 2\eta}{4} \right).$$

The bound controls all such smooth alternatives with $N \leq 1$, not just the witness in Proposition 13. At fixed targets their rescaled ambiguity is $O(m^2)$. For physical fields divide by $a(\eta) > 0$; uniform physical-field control requires staying away from the singular endpoints. Without a norm bound any nonzero hidden solution can instead be scaled to an arbitrarily large target difference. Neither convention is an observation model supplied by the geometry.

4.2 Errors in records are a different question

Equation (3) retains exact agreement on the whole past while changing the law. It is not a stability theorem for noisy records under a fixed law. Even at a massless-fixed target, an aggregate field-error norm need not control a point value. To see the regularity issue explicitly, fix any interior target (η_*, Y) and let Z_k be the real unit-L^2 zonal harmonic centered at Y, with $Z_k(Y) = k/\sqrt{2\pi^2}$. The smooth massless solutions

$$\psi_K(\eta, X) = \frac{1}{\sqrt{K}} \sum_{k=K}^{2K-1} \frac{\cos(k(\eta - \eta_*))}{k} Z_k(X)$$

have conserved conformal norm $N = 1$ and target value $\sqrt{K/(2\pi^2)}$, but

$$\|\psi_K\|^2_{L^2((0,2\pi)\times S^3)} = \frac{\pi}{K} \sum_{k=K}^{2K-1} \frac{1}{k^2} \leq \frac{\pi}{K^2}.$$

Their error on $I_{\mathrm{s}}^{-}(b)$ is even smaller. They do not agree *exactly* there. This is unbounded point evaluation under a specified norm, already present with records of the entire strip, not a causal-horizon or detector failure. Every fixed spatial average with profile $\chi \in L^2(S^3)$ instead satisfies $|\langle \chi, \psi_K(\eta)\rangle| \leq \|\chi\|_2/K$. No general past-record stability theorem for such averages is asserted.

A common finite spectral cutoff changes the exact question too: every solution in that restricted massless class is a finite real-analytic sum, so agreement on any nonempty open region entails agreement everywhere. Its fixing set is all of M. Thus finite mode sums cannot establish exact past agreement with a nonzero future difference in the unrestricted smooth theory. These are classical regularity observations, not further priority claims or results about the physical measurement process.

5 Unreality and openness

Put together with the companion theorem, these results separate three properties of an event x relative to a standpoint b:

- *real from* b: x is in b's fragment, which a local becoming theory fixes to lie between $I_{\mathrm{s}}^{-}(b)$ and $J^{-}(b)$;
- *guaranteed from* b: $x \in \mathrm{Det}(b)$, so that every law with the domain-of-dependence property fixes x from what is real from b;
- *fixed from* b *under given laws*: $x \in F_{\mathcal{L}}(b)$.

Every real event is fixed under any continuous law with the domain-of-dependence property, and every guaranteed event is fixed under each such law. The standard models exhibit two geometric regimes: in the flat, open, radiation and de Sitter models nothing beyond the fragment is guaranteed; in the closed dust model after maximum expansion, and in the Einstein static universe, everything is. So:

1. *Unreality does not entail openness.* An event can be unreal from b and guaranteed from b: in the closed dust model, every future event is so from every event after maximum expansion. Under particular laws, events can be unreal and fixed without being guaranteed: the conformal scalar in the closed dust model before maximum expansion fixes the echo and part of b's timelike future near the crunch (Theorem 12), and the flow law of Remark 2 fixes all of b's timelike future even in Minkowski spacetime. These examples illustrate a compatibility claim made by deterministic growing-block views: an event can be absent from a standpoint's real fragment while its value is fixed under the stipulated laws and solution class. On our reading, Rakíc's sea-fight intuition belongs here [18]: at most, the outcome is fixed from the observer, not real from her.

2. *Determinism does not entail fixedness from each event.* For a law and admissible solution class with unique evolution from Cauchy data, those data fix the solution throughout spacetime; nevertheless, no single event's past contains a Cauchy surface in the flat, open, radiation or de Sitter models. For the conformal scalar in Minkowski spacetime and the flat universes, what is fixed from b is exactly $J^{-}(b)$ (Proposition 7): from each standpoint the future is unfixed as well as unreal. In the open, closed radiation and de Sitter models we have shown only that nothing beyond the past is guaranteed.

3. *How much of the unreal future is fixed depends on two things.* The causal structure of the spacetime decides what is guaranteed, and the guarantee can change with cosmic epoch: in the closed dust model it is everything after maximum expansion and nothing beyond the past before it. The fields decide what is fixed beyond the guarantee; under the comoving flow law above, the answer to "is the future fixed from here?" differs between two events of the same spacetime. The causal structure of the given spacetime settles the first; it does not settle the second.

These distinctions leave both major ontological replies available. A deterministic growing-block theorist can accept that an event is absent from the real fragment while its field values are fixed by the laws and the fragment, and regard the later addition of events as ontological openness. Whether such addition suffices for openness, or whether openness requires lawful alternatives or indeterminacy, is a philosophical and semantic question not decided here. An eternalist can accept exactly the same fixing relations while holding that all events exist. Neither the definition of $F_{\mathcal{L}}$ nor the calculations select between these interpretations.

The positive constraint is on an inference: unreality from b does not by itself secure lawful non-fixing from b, and lawful fixing does not establish existence in b's fragment. Proposition 13 sharpens the dependence on laws: even the same geometry and the same exact past-agreement region allow different fixing verdicts when an ordinary mass is added. But (3) shows why a discontinuous verdict about exact equality need not be a large change in bounded field differences. Practical prediction requires a measurement process, observable, error norm and tolerance beyond these definitions.

5.1 Which existing realm is held fixed?

The agreement base must also be distinguished from the ontology's entire existing realm. The present comparison fixes $I_{\mathrm{s}}^{-}(b)$; it does not thereby fix all spacelike events. For example, the bow-tie version of relativistic GBT admits the elsewhere region as well as the past and the standpoint [7, Chap. 9]. Incoming radiation outside an observer's past can therefore distinguish two such realms even when the observer's entire past agrees. A past-agreement counterexample is not, without further argument, a counterexample to determination by that larger realm.

A bounded consequence follows within the fixed-background framework already used here. Put

$$W_b = I^{-}(I^{+}(b)) \setminus J^{+}(b).$$

This is an open region: its events are outside the future cone of b and share a future with b. If two solutions of a law with the stated domain-of-dependence property agree on W_b, they agree at every event of $J^{+}(b)$. To see this, choose a Cauchy temporal function t with $t(b) = 0$ and an earlier level $\Sigma = \{t = -1\}$. Every past-inextendible causal curve from $z \in J^{+}(b)$ meets Σ at some $y \leq z$. Such a y lies outside $J^{+}(b)$ and shares a chronological future with b: extend z slightly timelike and use push-up. Thus $y \in W_b$, and the domain-of-dependence property gives agreement at z.

This is sufficiency, not a claim that every point of W_b is necessary physical data. For continuous fields on a fixed common manifold, exact agreement on a dense subset already gives agreement on W_b. A separate causal-curve screening criterion may have a set-theoretically minimal realm without thereby finding a minimal physically determining data set. Nor does this argument compare two dynamical geometries: the background, operator and admissible global solution class remain fixed.

The larger comparison reinforces the distinction at issue. Under the named dynamics, a realm may determine future field values while excluding future events. Whether that exclusion is compatible with bivalent future truth or with later grounding remains a further question. Correia

and Rosenkranz's bivalent account explicitly distinguishes future truth from predetermination [7, Chap. 7]; the present calculation does not refute that account.

Three further limits matter. First, the comparison is conditional on the specified background and global solution class; the laws and that class are not derived from facts available in a past fragment. A fragmentalist may propose that standpoint-relative facts ground the tenseless structure rather than conversely. Without an independently defended explanatory asymmetry, that proposal adds no established truth-conditional or empirical consequence to the fixing relations. Second, the closed dust results concern an idealized model with linear classical fields on a fixed background; no claim about the actual cosmology or gravitational backreaction follows. Third, Manchak shows that *genuine* prediction, required to hold in every inextendible spacetime into which the observer's causal past embeds isometrically, reaches no further than the boundary of that past, even under any local conditions on matter [16, pp. 320–321]; $\mathrm{Det}(b)$ and $F_{\mathcal{L}}(b)$ are nomological notions given M, not epistemic ones.

6 A remark on standpoint logic

Read $\Box A$ at b as "at every event $x \leq b$, A" on the frame (M, R) with bRx iff $x \leq b$. If A is a tenseless claim about what occurs at x, this is the reflexive past modality of Diodorean tense logic, and the standpoint reading of Fine and Lipman [9, 15] adds no formal content. R is reflexive and transitive, so S4 is valid, and in Minkowski spacetime the valid formulas are exactly those of S4.2 [11]. The axiom .2, $\Diamond\Box A \to \Box\Diamond A$, is valid at a world for all valuations iff any two of its successors have a common successor [1]. So .2 is valid at b iff the subframe generated by b is directed, that is, iff any two events in $J^-(b)$ have a common causal past.

In a model conformal to $(\eta_-, \eta_+)\times\mathbb{R}^3$, two events have a common past iff $|X-Y| < \eta_x+\eta_y-2\eta_-$, so if $\eta_- > -\infty$ the directedness fails in the past of every event and .2 is valid at no event. This is the particle-horizon structure of Rindler [19] in modal form, and it holds equally with or without an inflationary era of finite conformal duration. The horizon problem of cosmology [13] is therefore not the failure of .2, which occurs with and without inflation, but the failure of particular instances: pairs of points on our last-scattering surface, at comoving separation up to $2(\eta_0 - \eta_{\mathrm{ls}})$, have no common past unless $\eta_0 - \eta_{\mathrm{ls}} < \eta_{\mathrm{ls}} - \eta_-$, which fails without inflation and holds with enough of it. Two cautions. If persistence is imposed on suitable tenseless past-event facts, rather than on centered claims such as "this event is present now," the corresponding valuations are constrained along the future direction, and the frame property that matters may concern common futures, which these models have when $\eta_+ = \infty$; the logic of persistent valuations is a separate question. And Lewis and van der Schaaf study causal modal logics of general spacetimes [24]; we have not checked how their results bear on the past-directed frames used here. Whether S4 is exactly the logic of all past-directed globally hyperbolic frames we leave open.

7 Conclusion

The unreality of the future from an event, which local theories of relativistic becoming deliver, neither implies nor excludes its being fixed from that event. What the causal structure guarantees, that is, what every local deterministic law fixes by the domain-of-dependence property alone, is characterized by a slicing condition on Geroch's domain of prediction (Theorem 3) and computable from conformal diagrams; it is a lower bound on what any particular law fixes. In most standard models nothing beyond an event's past is guaranteed, and in Minkowski spacetime and the flat universes a conformal scalar field shows that nothing more need be fixed; in the closed dust model

everything is guaranteed from every event after maximum expansion. Particular fields can fix more than is guaranteed: in the closed dust model before maximum expansion, a conformal scalar is fixed on part of an event's far future while its nearer future stays open, and with a preferred flow a field is fixed everywhere even in Minkowski spacetime. An arbitrarily small ordinary mass on the same dust background can remove fixing at a selected future event, while normalized hidden field differences there remain $O(m^2)$. Exact nonfixing, practical unpredictability and ontic addition are therefore separate questions. The question whether the future is open from a standpoint thus divides into a question about the causal structure of the spacetime, which settles the guarantee, and a question about the fields, which that structure does not settle.

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